P10FG-xxxxE/Z2:1(H30)LF

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PMB-SERIES Rev.04-2008

- ✓ 2 Watt
- ✓ **2:1** Wide Input
- ✓ **Reg. Single and Dual** Output
- ✓ **1 - 3 kV** DC I/O Isolation
- ✓ **DIP16** case
- ✓ Contin. Short Circuit Protection

The PMB 2Watt series is a family of cost effective 2W single and dual output DC/DC converters with an optional control Pin (SIP). These converters are encapsulated in an ultra miniature SIP8 or DIP16 plastic or metal case. High performance features: continuous / long time short circuit protection with automatic restart and tight line / load regulation, high efficiency operation and output voltage accuracy of ±2% maximum.

All specifications typical at Ta=25℃, nominal input voltage and full load unless otherwise specified

Input Specifications	
Voltage Range	2:1 Wide Input (see table)
Input Filter	Capacitor
Input Reflected Ripple Current[1]	35 mA pk-pk
Output Specifications	
Voltage Accuracy	± 2%
Short Circuit Protection	Indefinite (Automatic Recovery)
Line Regulation	± 0.5%
Load Regulation (25% - 100%)	± 1%
Ripple and Noise (20Mhz bandwidth)	80 mV pk-pk
Temperature Coefficient	± 0.02% / ℃
General Specifications	
I/O Isolation Voltage (3 sec.)	1000 VDC (3000 VDC optional)*
I/O Isolation Capacity	60 pF
I/O Isolation Resistance	1000 M Ohm
Switching Frequency	100 - 650 kHz
Humidity	95% rel H
Reliability Calculated MTBF (MIL-HDBK-217F)	> 2.732 Mhrs
Physical Specifications	
Case Material	Non Conductive Black Plastic (UL94V-0 rated)
Potting Material	Epoxy (UL94V-0 rated)
Weight	~ 6 g, typ.
Environment Specifications	
Operating Temperature	-40 to +85℃ (ambient)
Maximum Case Temperature	100℃
Storage Temperature	-40 to +125℃
Cooling	Free Air Convection (10mm distance required)
RoHS Conform	Soldering 260℃, max. (1.5mm from case 10s.)

Selection Guide
Single Output

Order #	Input Voltage (VDC)	Input Current No Load (mA)	Input Current Full Load (mA)	Output Voltage (VDC)	Output Current Min. Load (mA)	Output Current Full Load (mA)	Efficiency (%)	Capacitor Load (uF)[2]
SINGLE OUTPUT								
P10FG-053R3E2:1LF	4.5-9	15	492	3.3	125	500	67	3300
P10FG-0505E2:1LF	4.5-9	15	571	5	100	400	70	3300
P10FG-0509E2:1LF	4.5-9	30	555	9	56	222	72	470
P10FG-0512E2:1LF	4.5-9	30	555	12	42	167	72	470
P10FG-0515E2:1LF	4.5-9	30	547	15	33	133	73	470
P10FG-0524E2:1LF	4.5-9	60	533	24	21	83	75	220
P10FG-123R3E2:1LF	9-18	15	205	3.3	125	500	67	3300
P10FG-1205E2:1LF	9-18	15	216	5	100	400	77	3300
P10FG-1209E2:1LF	9-18	15	213	9	56	222	78	470
P10FG-1212E2:1LF	9-18	15	208	12	42	167	80	470
P10FG-1215E2:1LF	9-18	15	213	15	33	133	78	470
P10FG-1224E2:1LF	9-18	15	208	24	21	83	80	220
P10FG-243R3E2:1LF	18-36	8	98	3.3	125	500	70	3300
P10FG-2405E2:1LF	18-36	8	108	5	100	400	77	3300
P10FG-2409E2:1LF	18-36	8	104	9	56	222	80	470
P10FG-2412E2:1LF	18-36	8	104	12	42	167	80	470
P10FG-2415E2:1LF	18-36	8	104	15	33	133	80	470
P10FG-2424E2:1LF	18-36	8	104	24	21	83	80	220
P10FG-483R3E2:1LF	36-72	6	48	3.3	125	500	71	3300
P10FG-4805E2:1LF	36-72	6	56	5	100	400	74	3300
P10FG-4809E2:1LF	36-72	6	53	9	56	222	78	470
P10FG-4812E2:1LF	36-72	6	53	12	42	167	78	470
P10FG-4815E2:1LF	36-72	6	53	15	33	133	78	470
P10FG-4824E2:1LF	36-72	6	52	24	21	83	80	220

If you need other specifications, please enquire.

***OPTIONS:**

3 kV I/O Isolation	For optional 3kV DC I/O Isolation, please add “H30” before LF! (P10FG-2424E2:1**H30**LF for 3kV)

Selection Guide
Dual Output

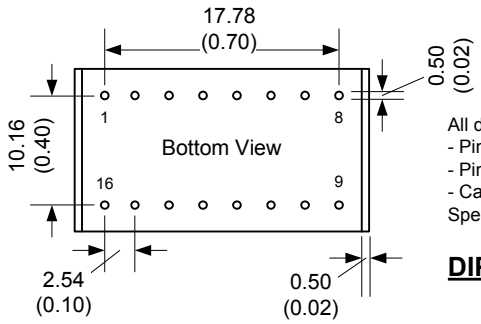
Order #	Input Voltage (VDC)	Input Current No Load (mA)	Input Current Full Load (mA)	Output Voltage (VDC)	Output Current Min. Load (mA)	Output Current Full Load (mA)	Efficiency (%)	Capacitor Load (uF)[2]
DUAL OUTPUT								
P10FG-053R3Z2:1LF	4.5-9	20	471	± 3.3	± 63	± 250	70	± 1000
P10FG-0505Z2:1LF	4.5-9	20	571	± 5.0	± 50	± 200	70	± 1000
P10FG-0509Z2:1LF	4.5-9	20	540	± 9.0	± 28	± 111	74	± 220
P10FG-0512Z2:1LF	4.5-9	25	533	± 12.0	± 21	± 83	75	± 220
P10FG-0515Z2:1LF	4.5-9	25	533	± 15.0	± 17	± 67	75	± 220
P10FG-0524Z2:1LF	4.5-9	60	563	± 24.0	± 10	± 42	71	± 100
P10FG-123R3Z2:1LF	9-18	15	188	± 3.3	± 63	± 250	73	± 1000
P10FG-1205Z2:1LF	9-18	15	222	± 5.0	± 50	± 200	75	± 1000
P10FG-1209Z2:1LF	9-18	15	210	± 9.0	± 28	± 111	79	± 220
P10FG-1212Z2:1LF	9-18	15	208	± 12.0	± 21	± 83	80	± 220
P10FG-1215Z2:1LF	9-18	15	210	± 15.0	± 17	± 67	79	± 220
P10FG-1224Z2:1LF	9-18	30	219	± 24.0	± 10	± 42	76	± 100
P10FG-243R3Z2:1LF	18-36	8	94	± 3.3	± 63	± 250	73	± 1000
P10FG-2405Z2:1LF	18-36	8	106	± 5.0	± 50	± 200	78	± 1000
P10FG-2409Z2:1LF	18-36	8	105	± 9.0	± 28	± 111	79	± 220
P10FG-2412Z2:1LF	18-36	8	104	± 12.0	± 21	± 83	80	± 220
P10FG-2415Z2:1LF	18-36	8	104	± 15.0	± 17	± 67	80	± 220
P10FG-2424Z2:1LF	18-36	20	106	± 24.0	± 10	± 42	78	± 100
P10FG-483R3Z2:1LF	36-72	6	47	± 3.3	± 63	± 250	73	± 1000
P10FG-4805Z2:1LF	36-72	6	56	± 5.0	± 50	± 200	74	± 1000
P10FG-4809Z2:1LF	36-72	6	53	± 9.0	± 28	± 111	79	± 220
P10FG-4812Z2:1LF	36-72	6	53	± 12.0	± 21	± 83	79	± 220
P10FG-4815Z2:1LF	36-72	6	52	± 15.0	± 17	± 67	80	± 220
P10FG-4824Z2:1LF	36-72	12	55	± 24.0	± 10	± 42	75	± 100

If you need other specifications, please enquire.

***OPTIONS:**

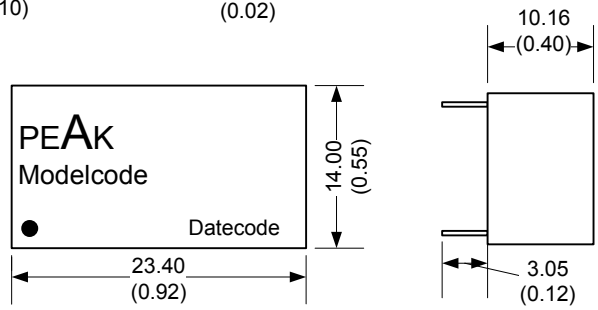
3 kV I/O Isolation	For optional 3kV DC I/O Isolation, please add “H30” before LF! (P10FG-2424Z2:1**H30**LF for 3kV)

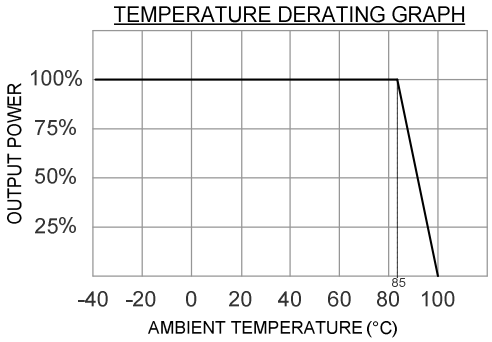
Package / Pinning / Derating



All dimensions are typical in millimeters (inches).
- Pin diameter: 1.0 +/-0.05 (0.04 +/-0.002)
- Pin pitch tolerance: +/-0.35 (+/-0.014)
- Case tolerance +/-0.5 (+/-0.02)
Specification may change without notice.

DIP16 – PLASTIC CASE





PIN CONNECTIONS		
#	SINGLE	DUAL
1	- Vin	- Vin
2	- Vin	- Vin
6	N.C.	Common
8	N.C.	- Vout
9	+Vout	+Vout
11	- Vout	Common
15	+Vin	+Vin
16	+Vin	+Vin

(same pinning for 3kV)

App Notes:

[1] = Measured Input reflected ripple current with a simulated source inductance of 12uH
[2] = Tested by nominal Vin and constant resistive load.

- Operation under no-load conditions will not damage these devices, but they will not observe the listed specifications.